

Ref – Energy Calculated Spread sheet.

Power Generation						
Power Source	Voltage [V]	Current [I]	% of Generation/day	Required Power	Power [P]- KW/hr	Power/day
Solar	430	16.7	60	72	43	1032
ESS	480	16.7	30	80	24	576
Grid	435	20.0	10	87	9	209
Total Power Generation per day in KW/24 hr						1817

PCC - Point of common coupling		
PCC will Logically controlled on availability of source		
Voltage - [V]	430	Output Power/day in KW
Current - [I]	17	
Power - [W] - KW/h	75	1800 Gen /day
Power Availability	1.17	1772 use / day

ESS - Charging	
No. of batteries	40
Voltage	12
Amper hour - capacity	100
Power / charge - Watt / hr	48000
Power / one cycle charge - KW	48

Fast Charging Station Load							
Cars	Voltage (V)	Power (kW)	Battery Capacity (kVh)	SOC Available	Required Current	Charging Time in hr	Unit charge Cost
1- Tata Nexon - EV	400	30.2	30.2	20	75.5	2.47	10 745.83
2- MG ZS EV	400	50	44.5	30	125	1.49	10 743.75
3-Tata Trigo EV	400	16.8	26	50	42	4.40	10 739.58
4- Mahindra eV	400	70	60	10	175	1.07	10 747.92
Total Power		167			Total Time	9	Total Revenue 2977
Total Power		1670	KW power can charge / day		Total Time	/ 96 hr	Total Revenue 26794

Battery Swapping Station	
Total Batteries	30
Under maintenance	0
Available batteries	30
Voltage	12
Current	10
Time / Battery in hr	5
No of Cycle per day	3
Power/Battery - KW	0.12
Total Station KW	18
Total Energy Required for Full Swapping Station per Day - KW	54

Total Batteries charging /	90
Cost of charge per batter	30
Total Amount Per day	2700

Total Energy claculation				
Power sources	KW / day	KW / Month	KW / year	KW / 5 years
Required full Power From Grid	1,817	54,510	663,205	3,316,025
Solar	1,032	30,960	376,680	1,883,400
ESS	576	17,280	210,240	1,051,200
Grid	209	6,270	76,285	381,425
Total Power saving / Day	1,608	48,240	586,920	2,934,600
Total Generation Cost / Day 12rs / unit	21,804	654,120	7,958,460	39,792,300
Total Generation Cost Saving / Day 12rs / unit	19,296	578,880	7,043,040	35,215,200
Spending Amount	2,508	75,240	915,420	4,577,100

Revenue				
Source	Per day	Per Month	Per Year	Per 5 Years
Fast charging	26,794	803,813	9,779,719	48,898,594
Battery Swapping	2,700	81,000	985,500	4,927,500
Total Revenue	29,494	884,813	10,765,219	53,826,094
Spending Amount	2,508	75,240	915,420	4,577,100
Total Profit	26,986	809,573	9,849,799	49,248,994